A RATIONAL SURFACE SINGULARITY WITH A DUALIZING COMPLEX IS MADE GORENSTEIN BY BLOWUPS IN SINGULAR CLOSED POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the construction of a blowup making the strict transform of the dualizing module invertible, its domination by a sequence of normalized blowups at singular closed points with the normality of every term, and the pointwise verification of invertibility at regular and singular points closed by the torsion freeness of the dualizing module.

Editorial note. The label lemma-rational-to-gorenstein names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-rational-to-gorenstein –start 2687 –end 2743 –local-id 28081 returned RESULT=OK (L28081 free, occupancy FREE, public interval clear). Pre-assembly tag self-check: canonical tag resolved and not present in reference/public-items.jsonl.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Invertibility of the strict transform of the dualizing module. Divisors, the lemma producing a blowup with respect to the Fitting ideal such that the strict transform of a given coherent module is invertible. Supplies the blowup over which the strict transform of the dualizing module is an invertible sheaf.
- Domination by normalized blowups and normality of the blowups. Same chapter, the lemma dominating a proper birational morphism by normalized blowups at closed points, and the lemma that the blowup of the maximal ideal of a normal rational surface singularity is normal. Supply the sequence of normalized blowups dominating the chosen blowup together with the fact that each step is an ordinary blowup and every term is normal.
- Dualizing modules of regular and of blown up rational singularities, and injectivity of the comparison map. Dualizing Complexes, the lemma that a regular scheme is Gorenstein, and Same chapter, the lemmas describing the dualizing module of the blowup and the surjectivity of the comparison

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map along such a blowup. Supply the invertibility at regular points and the surjectivity on stalks along the sequence at singular points.
- Torsion freeness of the dualizing module. Duality for Schemes, the lemma that the dualizing module of a normal scheme is torsion free. Supplies the final identification of the quotient by torsion with the dualizing module of the last term.

Source reference. Original lemma 0ESN: sources/stacks-chow/tags.txt, source lines 18488–18488. Prerequisite pointer invoked at proof line(s) 2708; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AWK: sources/stacks-chow/tags.txt, source lines 13723–13723. Prerequisite pointer invoked at proof line(s) 2740; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AWX: sources/stacks-chow/tags.txt, source lines 13735–13735. Prerequisite pointer invoked at proof line(s) 2723; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B4Z: sources/stacks-resolve/pinned-original-source.tex, source lines 2534–2534. Prerequisite pointer invoked at proof line(s) 2714; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AGR: sources/stacks-resolve/pinned-original-source.tex, source lines 727–727. Prerequisite pointer invoked at proof line(s) 2726; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BBT: sources/stacks-resolve/pinned-original-source.tex, source lines 1277–1277. Prerequisite pointer invoked at proof line(s) 2709; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B64: sources/stacks-resolve/pinned-original-source.tex, source lines 2641–2641. Prerequisite pointer invoked at proof line(s) 2731; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0BBU: sources/stacks-resolve/pinned-original-source.tex, source lines 2618–2618. Prerequisite pointer invoked at proof line(s) 2728; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa)$ be a local normal Nagata domain of dimension 2 which defines a rational singularity. Assume A has a dualizing complex. Then there exists a finite sequence of blowups in singular closed points*

$$X = X_n \to X_{n-1} \to \ldots \to X_1 \to X_0 = \mathrm{Spec}(A)$$

such that X_i is normal for each i and such that the dualizing sheaf ω_X of X is an invertible $\mathcal{O}_X$-module.

Proof. The dualizing module ω_A is a finite A-module whose stalk at the generic point is invertible. Namely, $\omega_A \otimes_A K$ is a dualizing module for the fraction field K of A, hence has rank 1. Thus there exists a blowup $b : Y \to \mathrm{Spec}(A)$ such that the strict transform of ω_A with respect to b is an invertible $\mathcal{O}_Y$-module, see Divisors, Lemma 0ESN. By Lemma 0BBT we can choose a sequence of normalized blowups

$$X_n \to X_{n-1} \to \ldots \to X_1 \to \mathrm{Spec}(A)$$

such that X_n dominates Y. By Lemma 0B4Z and arguing by induction each $X_i \to X_{i-1}$ is simply a blowing up.

We claim that ω_{X_n} is invertible. Since ω_{X_n} is a coherent $\mathcal{O}_{X_n}$-module, it suffices to see its stalks are invertible modules. If $x \in X_n$ is a regular point, then this is clear from the fact that regular schemes are Gorenstein (Dualizing Complexes, Lemma 0AWX). If x is a singular point of X_n, then each of the images $x_i \in X_i$ of x is a singular point (because the blowup of a regular point is regular by Lemma 0AGR). Consider the canonical map $f_n^*\omega_A \to \omega_{X_n}$ of Lemma 0BBU. For each i the morphism $X_{i+1} \to X_i$ is either a blowup of x_i or an isomorphism at x_i. Since x_i is always a singular point, it follows from Lemma 0B64 and induction that the maps $f_i^*\omega_A \to \omega_{X_i}$ is always surjective on stalks at x_i. Hence

$$(f_n^*\omega_A)_x \longrightarrow \omega_{X_n,x}$$

is surjective. On the other hand, by our choice of b the quotient of $f_n^*\omega_A$ by its torsion submodule is an invertible module $\mathcal{L}$. Moreover, the dualizing module is torsion free (Duality for Schemes, Lemma 0AWK). It follows that $\mathcal{L}_x \cong \omega_{X_n,x}$ and the proof is complete. □